

Final Presentation

Credit Risk Assessment

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Idea & Description

↗ Project Idea & Description

The Lending Challenge

What is Lending?

A practice spanning hundreds of years where a **lender** with spare money provides funds to a **borrower** who needs it. The borrower repays the principal plus **interest** as a service fee.

The Critical Question

Since borrowers can potentially go bankrupt and fail to repay, **careful decision-making is crucial** for lenders to ensure they don't lose money.

Risk for Lenders

Avoid losses from non-repayment

Protection for Borrowers

Avoid unmanageable debt

Lending 101

Evolution of Lending Decisions

From personal relationships to machine learning



1800s

Relationship-Based

Personal knowledge, trade references, and blacklists



1950s

Statistical Scorecards

FICO credit scores using financial indicators



1970s

Regulatory Framework

FCRA and ECOA govern lending decisions



Today

Machine Learning

ML models for better credit ranking

Today's Major Principles

1

Allowed data variables only

2

Strict data collection & sharing

3

Decision interpretability

4

Nationwide consistency

ML Stack



Core Objective

Train an ML model to assess whether a **business is likely to repay a loan** using XGBoost classification with high accuracy on validation sets without overfitting.



Input Data

Business financial indicators: trades in past 90 days, max delinquency, external risk estimate, and more



XGBoost Model

Machine learning algorithm trained to classify businesses into good/bad borrower categories



Output

Binary classification: likely to repay (good borrower) or unlikely to repay (bad borrower)

Why This Matters

• For Lenders

Avoid financial losses by identifying high-risk borrowers before lending

• For Borrowers

Avoid taking on debt they may not be in the best position to repay

Approach - Feature Modifications

- Replaced FICO special codes with NaN and kept raw numeric features.
- Avoided scaling since XGBoost splits by rank, not magnitude.
- Flipped target so **Bad = 1** for proper class-weight handling.
- Applied `scale_pos_weight` to emphasize detecting high-risk borrowers.

Approach - Param Search

- Ran wide hyperparameter search to identify stable tree structure (Used `RandomizedSearchCV` to narrow down the best values).

```
1 param_dist = {  
2     "max_depth": [3, 4, 5],  
3     "min_child_weight": [2, 4, 6, 10],  
4     "learning_rate": [0.0075, 0.01, 0.015, 0.02],  
5     "n_estimators": [600, 900, 1200, 1500],  
6     "subsample": [0.6, 0.75, 0.9],  
7     "colsample_bytree": [0.6, 0.75, 0.9],  
8     "reg_alpha": [0, 1],  
9     "reg_lambda": [0, 1],  
10    "gamma": [0, 0.1, 0.3, 0.5],  
11 }
```

- Applied threshold sweeping to optimize Bad recall and F1.
- Selected threshold ~ 0.40 to maximize practical risk detection.

Approach - Monotonicity

- Added monotonic constraints to enforce economically logical trends.
- Used Spearman correlations to derive data-driven monotone directions.
- Removed weak/unstable constraints to avoid over-restriction.
- Final model achieves strong Bad recall, calibrated probabilities, and interpretable behavior.

Performance Metrics

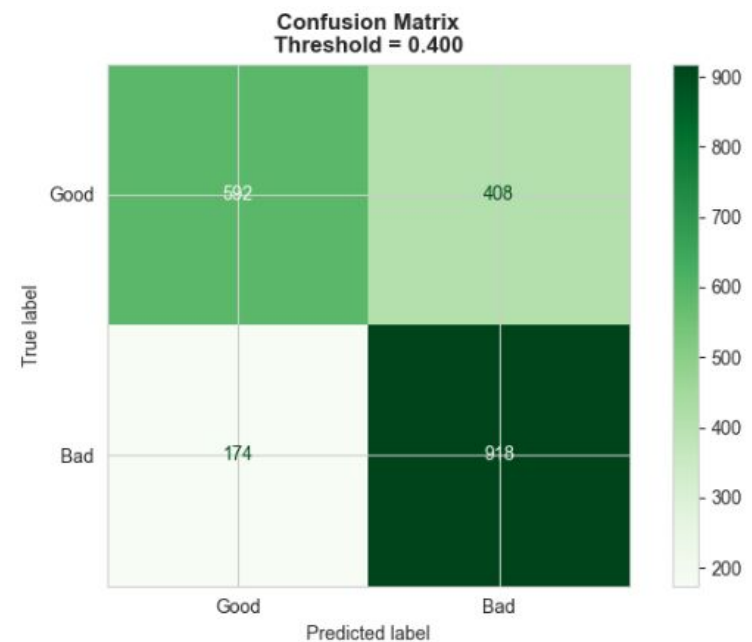
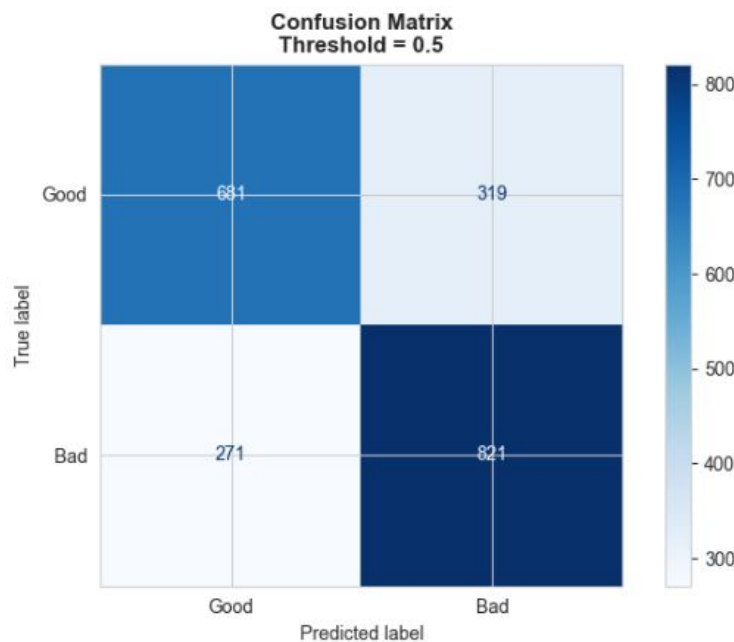
- ROC–AUC 0.79 – good class separation; model can distinguish Good vs Bad borrowers
- Recall (Bad) 0.84 – catches most risky borrowers; fewer “Bad” customers slip through
- Precision (Bad) 0.69 – some false alarms; but acceptable for safety-focused lending
- F1 (Bad) 0.76 – balanced recall & precision; stable model performance
- Accuracy 0.72 – overall correct predictions; secondary metric
- AP ~0.79 – strong performance even with class imbalance

FINAL PERFORMANCE ASSESSMENT

	Objective	Target	Achieved	Status	Margin
	ROC-AUC ≥ 0.75	0.750	0.792479	PASS	0.042479
	Recall (Bad) ≥ 0.70	0.700	0.840659	PASS	0.140659
	Precision (Bad) ≥ 0.65	0.650	0.692308	PASS	0.042308
	F1-Score ≥ 0.67	0.670	0.759305	PASS	0.089305
	Beat Random Baseline (0.50 AUC)	0.500	0.792479	PASS	0.292479
	Beat Majority Baseline (52.2% Acc)	0.522	0.721797	PASS	0.199797

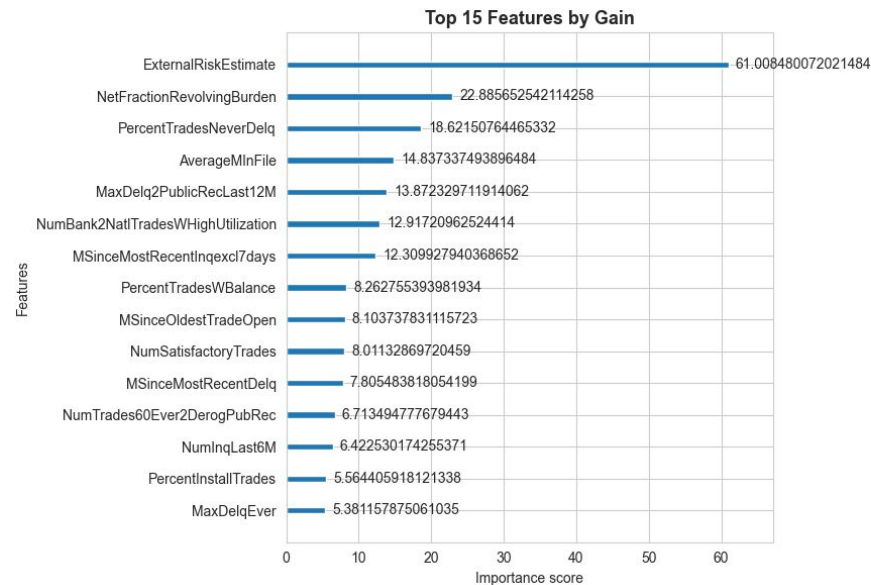
Threshold & Confusion Matrices

- Best threshold 0.40 – best recall/F1 balance
- False negatives drop from 271 to 174 – fewer missed Bad borrowers
- More false positives – some Good borrowers flagged as Bad
- Trade-off acceptable – safer to avoid risky loans
- 0.40 works better for lending perspective



Evaluations

- Monotonic constraints keep predictions logical
- Feature importance aligns with credit-risk theory
- ExternalRiskEstimate = strongest indicator
- Model decisions are easy to explain
- Reliable for lenders – consistent, interpretable, risk-aware model
- Small trade-off: stricter approval for Good borrowers



Future Work

- Expand the dataset with more borrower records to improve model stability and reduce noise in the predictions.
- Improve data quality by reducing the number of entries with missing, unverified or unavailable borrower information.
- Use larger compute resources to run a wider and deeper hyperparameter search for XGBoost.
- Explore extreme parameter ranges since several current parameters reached boundary values, suggesting better configurations may exist outside the tested space.
- Revisit the false-negative tradeoff with richer data to better balance fairness to good borrowers and risk control for lenders.

Conclusion

Q&A